Rutwik Kadam

rutwik2602@gmail.com | 9284908856 | LinkedIn | GitHub

Profile Summary

Generative AI Developer and Machine Learning Engineer specializing in building Generative AI solutions for FinTech and EdTech startups. Proven track record of reducing infrastructure costs by 90% through Edge AI solutions and optimizing data latency by 40%. Highly proficient in Python, Machine Learning algorithms, and FastAPI, with hands-on experience in LLM orchestration using foundational model, vector databases and real-time Voice AI assistants.

Education

Masters in Computer Science, Specialization in AI — MIT — Aug 2023 – Jul 2025
CGPA: 8.73/10

Bachelors in Computer Science — Modern College of Science, Pune — Jul 2020 – May 2023
CGPA: 9.22/10

Technical Skills

Programming Languages: Python, JavaScript.

AI/ML & Data: PyTorch, NumPy, Pandas, Transformers, Deep Learning, Regression & Classification, Model Evaluation, RNN, CNN, LSTM, Transformer Architecture.

Generative AI: Prompt Engineering, **RAG** (Hybrid Retrieval, ColBERT), LoRA, PEFT, Quantization, Encoder–Decod Architectures, Mistral, DeepSeek, Agentic AI (CrewAI), **Retrieval Augmented Generation Concepts**

LLM orchestration: LangChain, LangGraph, CrewAI , AutoGen, Semantic Kernel, DSPy, Flowise, LangFlow, OpenAI Swarm, PydanticAI

Backend & APIs: FastAPI, REST APIs, WebSockets, FastRTC

Databases: Vector Databases (Qdrant, ChromaDB), MongoDB, PGVector

Cloud & DevOps: AWS, Docker, GitHub Actions, CI/CD, MLOps, Hugging Face (Inference Endpoints)

Work Experience

NG AI labs - AI Platform Engineer — Aug 2025 – Present

Speech - to - Speech Model - link

- **Trained a production** - grade, end - to - end Speech → Text → Speech model in English, Hindi, Gujarati, Indian English accent and deployed across multiple products including an AI Interviewer and Voice Assistant.
- Trained on 50+ hours of audio and achieved <10ms speech synthesis latency with a Word Error Rate (WER) of 10%.
- Enabled voice cloning with fewer than 20 audio samples, making it highly data - efficient.
- The quantized model supports deployment on low end / edge devices.
- Reduced speech synthesis **cost from $18/hr → $0.12/hr** (99% cost reduction).
- **Tech Stack**: Python, PyTorch, ONNX, Typescript for inferencing.
- **Deployment**: HuggingFace, Sagemaker, Azure.

Zoe AI Interviewer - link

- **Built an AI - powered mock interview platform** that conducts interviews, evaluates responses, and provides personalized post - interview feedback.
- **Designed a RAG pipeline using Qdrant** to retrieve context from a user's past interview history, enabling the AI to identify weak areas and ask targeted follow - up questions.
- Optimized the system flow to under 1 second, supporting 10,000 concurrent interviews with an end - to - end STS latency of <3 seconds - outperforming OpenAI, Google Gemini, and ElevenLabs.
- Orchestrated multi - step AI agent workflows using LangGraph for robust state, memory, and context manage-ment across interview sessions.
- **Tech Stack**: Python, FastAPI, LangGraph, Pandas, React.
- **Deployment and Database**: AWS EC2, Amplify, Qdrant, Firebase.

InternetSoft – GEN AI Developer — Jan 2025 – Jul 2025

Multi - Agent Testing System

- Engineered a **multi - agent QA automation system** that autonomously tested a POS platform, detected bugs, and reported them directly to Zoho - eliminating manual reporting.
- Designed a 3 - stage agentic orchestration pipeline.
- Assigner Agent - analyzed bug severity & developer expertise to route issues optimally.
- Reviewer Agent - triggered on GitHub commits to validate fixes before merge.
- Production Agent - performed final cross - validation of fix, review report & code diff before deployment.
- Reduced manual QA intervention by 60%, accelerating the end - to - end development cycle.
- **Tech Stack**: Python, LangChain, LangGraph, RAG, Foundation Model.

Projects

Lammatization for Local language - MIT

- Trained a model to convert Marathi words to their base forms using the Hidden Markov Model (HMM) algorithm.
- Successfully achieved 81.3%
- Developed an algorithm capable of processing 1 GB of Marathi textual data to reduce token size and perform normalization efficiently.
- The model was further utilized by AI4Bharat for corpus normalization.
- Reduced normalization cost and time by 70%.